

Q: Given an ADC with 16-bit resolution and ± 1 V range, what is the voltage resolution?

A AI Model: Gemma4

Q: Given an ADC with 16-bit resolution and ± 1 V range, what is the voltage resolution?

Time: (1m 46.28s)

A:

The voltage resolution (ΔV) is determined by the inherent quantization step size, which represents the smallest voltage increment the ADC can distinguish between output codes relative to its total input range (V_{FS}). For an ideal N -bit ADC operating over the full-scale range (FSR), this resolution is defined mathematically as $\Delta V = \frac{V_{FS}}{2^N}$. Given a 16-bit resolution ($N = 16$) and a symmetrical range of ± 1 V, the total Full-Scale Range is $V_{FS} = 1 \text{ V} - (-1 \text{ V}) = 2 \text{ V}$. Substituting these values yields $\Delta V = \frac{2 \text{ V}}{2^{16}} = \frac{2 \text{ V}}{65,536}$, resulting in a theoretical voltage resolution of approximately $30.5 \mu\text{V}$ (30.5 microvolts). This calculation assumes perfect linearity and quantization noise equivalent to $\text{LSB}_{\text{noise}} = 1 \text{ LSB}$. In practice, the actual measurable resolution can be degraded by finite op-amp gain, non-linearities (INL), comparator offset voltage, or reference inaccuracies.

B AI Model: Llama3.1

Q: Given an ADC with 16-bit resolution and ± 1 V range, what is the voltage resolution?

Time: (32.33s)

A:

The voltage resolution (ΔV) can be calculated as:

$$\Delta V = V_{\text{ref}} / 2^N,$$

where V_{ref} is the reference voltage (± 1 V in this case), and N is the number of bits (16).

First, we need to calculate the total number of possible digital output values:

$$2^{16} = 65536$$

Since the ADC has a ± 1 V range, we can use either half of the range as the reference voltage:

$V_{\text{ref}} = 1/2 = 0.5$ V (or -0.5 V for the negative range)

Now, substitute these values into the equation:

$$\Delta V = 0.5 \text{ V} / 65536 \approx 7.62 \times 10^{-6} \text{ V}$$

This is equivalent to approximately $76.2 \mu\text{V}$.

C AI Model: Qwen3.5

Q: Given an ADC with 16-bit resolution and ± 1 V range, what is the voltage resolution?

Time: (23m 13.76s)

A:

For a 16-bit ADC with a ± 1 V input range, the full-scale span (V_{FS}) extends from -1 V to $+1$ V, totaling 2 V. The ideal voltage resolution, representing the analog change corresponding to one digital Least Significant Bit (LSB), is derived by dividing this total dynamic range by the number of quantization intervals. While exact code counts may define steps as $(2^N - 1)$ for specific edge treatments, standard engineering practice approximates the LSB weight using $V_{FS}/2^N$. Applying this to the given specifications: $\Delta V = \frac{(+1 \text{ V}) - (-1 \text{ V})}{2^{16}} = \frac{2 \text{ V}}{65,536}$, yielding a resolution of approximately $30.5 \mu\text{V}$. This theoretical step size assumes ideal quantization without differential non-linearity (DNL) errors or missing codes; in practice, circuit noise and comparator offsets might render transitions smaller than this LSB indistinguishable from thermal noise at room temperature (kT/C considerations), effectively reducing the usable resolution below $30.5 \mu\text{V}$ under suboptimal signal-to-noise conditions.

D AI Model: Ministral-3

Q: Given an ADC with 16-bit resolution and ± 1 V range, what is the voltage resolution?

Time: (1m 7.28s)

A:

The **voltage resolution** (LSB size) of a 16-bit ADC spanning ± 1 V (± 2 V total range) is calculated as:

$$\text{LSB} = \frac{\text{Full-Scale Range}}{\text{Number of Quantization Levels}} = \frac{2 \text{ V}}{2^{16}} = \frac{2}{65,536} \approx 30.5176 \mu\text{V}$$

Key assumptions:

- **Unipolar vs bipolar:** The $\pm 1\text{ V}$ range implies a **bipolar input** ($\pm 1\text{ V}$), so the full-scale swing is **2 V total**, not 1 V (unipolar). If it were unipolar (0–1 V), the LSB would be **61.035 μV** .
- **Ideal quantization:** This assumes no missing codes or differential nonlinearity (DNL) errors; real ADCs may exhibit $\pm \frac{1}{2}$ LSB INL/DNL due to circuit mismatches.
- **Thermal noise:** The LSB size also sets the **noise floor** for signal-to-noise ratio (SNR) calculations, where $\text{SNR} \approx 6.02N + 1.76\text{ dB}$ for an ideal ADC ($N = 16$, $\text{SNR} \approx 98\text{ dB}$).

Practical note: In real ADCs, **offset errors**, **reference mismatches**, or **comparator hysteresis** can degrade effective resolution beyond the theoretical LSB. For example, a $\pm 2\text{ mV}$ offset would shift the zero crossing by ~ 65 LSBs ($\approx 2\text{ mV}/30\text{ }\mu\text{V} \approx 66$), requiring calibration..
